



Vulnerability Assessment &
Penetration Testing (VAPT) Report

4Africa Data Specialists

14 December 2023



TABLE OF CONTENTS

1	EXECUTIVE SUMMARY	3
2	SCOPE OF TESTING	3
3	METHODOLOGIES USED	3
3.1	NESSUS	3
3.2	GREENBONE ENTERPRISE	3
4	STAY ONE STEP AHEAD OF ATTACKERS	4
5	INFRASTRUCTURE.....	4
5.1	41.76.213.236.....	4
6	REPORTING CONCLUSION	7
7	AUDITORS	8



1 EXECUTIVE SUMMARY

The purpose of this assessment was to point out security loopholes, business logic errors, and missing best security practices. The tests were carried out assuming the identity of an attacker or a malicious user but, no harm was made to the functionality or working of the application / network. A full scan was also performed on the infrastructure to demonstrate a known attack, i.e. someone on the internal network or same public subnetted infrastructure. The scans and were performed over 6 (six) business days, as full and slow comprehensive audits were performed. These types of scans also look at unknown connectivity options, including packet SYN requests.

The overall results will be published below. Some recommendations however, is offered and also mentioned throughout this document.

2 SCOPE OF TESTING

Security assessments include testing for security loopholes in the scope defined below. Apart from the following, no other information was provided. Nothing was assumed at the start of the security assessment. The following scope of servers were covered under the security audit:

- Server hosted at Afrihost – 41.76.213.236

3 METHODOLOGIES USED

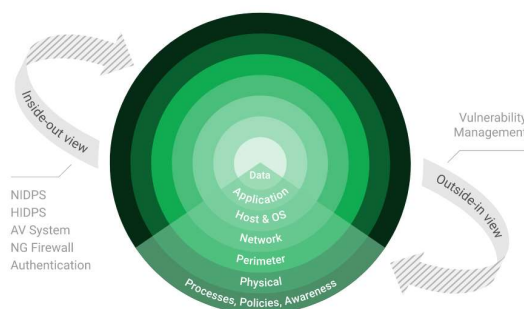
Sautech performed various audit techniques that includes the following:

3.1 Nessus

Nessus is a remote security scanning tool, which scans a computer and raises an alert if it discovers any vulnerabilities that malicious hackers could use to gain access to any computer you have connected to a network. It does this by running over 1200 checks on a given computer, testing to see if any of these attacks could be used to break into the computer or otherwise harm it.

3.2 Greenbone Enterprise

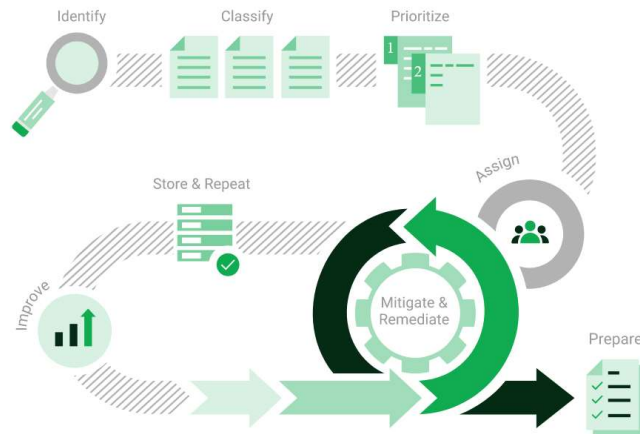
Greenbone is the world's most widely used vulnerability management provider.





4 STAY ONE STEP AHEAD OF ATTACKERS

IT security is a process – Open Source vulnerability management provides the foundation. Once you know where the chinks are in your armor, you can do something about them.



The process – from recognition to remedy and monitoring – represents a continuous cycle. You'll always be one step ahead of attackers.

5 INFRASTRUCTURE

5.1 41.76.213.236

We have performed three types of scans / attacks on the production server namely:

- Vulnerability Assessments.

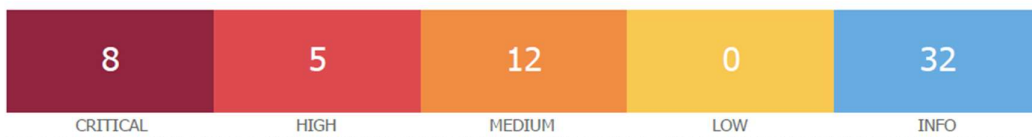
Vulnerability Assessment

Multiple technologies were used to perform a vulnerability assessment, below is the summarised version of the relevant results:

We have attached the below “highlight” in a separate report so that you can click on the “Plugin” link to get the resolution.



41.76.213.236



Vulnerabilities

Total: 57

SEVERITY	CVSS V3.0	VPR SCORE	PLUGIN	NAME
CRITICAL	9.8	7.4	161948	Apache 2.4.x < 2.4.54 Multiple Vulnerabilities
CRITICAL	9.8	9.0	172186	Apache 2.4.x < 2.4.56 Multiple Vulnerabilities
CRITICAL	9.8	7.4	160477	OpenSSL 1.1.1 < 1.1.1o Vulnerability
CRITICAL	9.8	7.4	162420	OpenSSL 1.1.1 < 1.1.1p Vulnerability
CRITICAL	9.8	6.7	166703	PHP 8.1.x < 8.1.12 Multiple Vulnerabilities
CRITICAL	9.8	6.7	179317	PHP 8.1.x < 8.1.22 Multiple Vulnerabilities
CRITICAL	9.8	6.7	162844	PHP 8.1.x < 8.1.8
CRITICAL	9.0	6.5	170113	Apache 2.4.x < 2.4.55 Multiple Vulnerabilities
HIGH	8.8	6.7	161975	PHP 8.1.x < 8.1.7 Multiple Vulnerabilities
HIGH	7.5	4.4	183391	Apache 2.4.x < 2.4.58 Multiple Vulnerabilities
HIGH	7.5	3.6	171465	PHP 8.1.x < 8.1.16
HIGH	7.4	6.0	171079	OpenSSL 1.1.1 < 1.1.1t Multiple Vulnerabilities
HIGH	N/A	6.3	169631	PHP 8.1.x < 8.1.14
MEDIUM	6.5	-	142960	HSTS Missing From HTTPS Server (RFC 6797)
MEDIUM	6.5	4.4	165551	PHP 8.1.x < 8.1.11 Multiple Vulnerabilities
MEDIUM	6.5	-	51192	SSL Certificate Cannot Be Trusted
MEDIUM	6.5	-	104743	TLS Version 1.0 Protocol Detection
MEDIUM	6.5	-	157288	TLS Version 1.1 Protocol Deprecated
MEDIUM	5.3	4.0	11213	HTTP TRACE / TRACK Methods Allowed



MEDIUM	5.3	2.9	162721	OpenSSL 1.1.1 < 1.1.1q Vulnerability
MEDIUM	5.3	4.4	173260	OpenSSL 1.1.1 < 1.1.1u Multiple Vulnerabilities
MEDIUM	5.3	2.9	178475	OpenSSL 1.1.1 < 1.1.1v Vulnerability
MEDIUM	5.3	2.9	184811	OpenSSL 1.1.1 < 1.1.1x Vulnerability
MEDIUM	5.3	-	15901	SSL Certificate Expiry
MEDIUM	4.3	1.4	177510	PHP 8.1.x < 8.1.20
INFO	N/A	-	10114	ICMP Timestamp Request Remote Date Disclosure
INFO	N/A	-	46180	Additional DNS Hostnames
INFO	N/A	-	48204	Apache HTTP Server Version
INFO	N/A	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	54615	Device Type
INFO	N/A	-	84502	HSTS Missing From HTTPS Server
INFO	N/A	-	43111	HTTP Methods Allowed (per directory)
INFO	N/A	-	10107	HTTP Server Type and Version
INFO	N/A	-	12053	Host Fully Qualified Domain Name (FQDN) Resolution
INFO	N/A	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	11219	Nessus SYN scanner
INFO	N/A	-	19506	Nessus Scan Information
INFO	N/A	-	11936	OS Identification
INFO	N/A	-	50845	OpenSSL Detection
INFO	N/A	-	57323	OpenSSL Version Detection
INFO	N/A	-	48243	PHP Version Detection
INFO	N/A	-	66334	Patch Report
INFO	N/A	-	26024	PostgreSQL Server Detection
INFO	N/A	-	56984	SSL / TLS Versions Supported



INFO	N/A	-	10863	SSL Certificate Information
INFO	N/A	-	70544	SSL Cipher Block Chaining Cipher Suites Supported
INFO	N/A	-	21643	SSL Cipher Suites Supported
INFO	N/A	-	57041	SSL Perfect Forward Secrecy Cipher Suites Supported
INFO	N/A	-	156899	SSL/TLS Recommended Cipher Suites
INFO	N/A	-	22964	Service Detection
INFO	N/A	-	25220	TCP/IP Timestamps Supported
INFO	N/A	-	84821	TLS ALPN Supported Protocol Enumeration
INFO	N/A	-	121010	TLS Version 1.1 Protocol Detection
INFO	N/A	-	136318	TLS Version 1.2 Protocol Detection
INFO	N/A	-	10287	Traceroute Information
INFO	N/A	-	10386	Web Server No 404 Error Code Check
INFO	N/A	-	10302	Web Server robots.txt Information Disclosure

* indicates the v3.0 score was not available; the v2.0 score is shown

6 REPORTING CONCLUSION

Based on the reports, it is best to attach these separate to the reporting document. These reports can then be shared with the relevant contractors and/or support personnel to implement the changes and updates.

The HTTPS/TLS updates as these are open to the public, low and medium risk, should be tested and implemented as soon as possible.

We do recommend that SentinelOne is implemented into the production environment as soon as possible.

Two addendum reports are attached to this report and should be read as one document.

With the following test performed:

- External HTTP and Port exploits
- Vulnerability assessment on CVE
- Penetration testing on login ports i.e.: ssh open



7 AUDITORS

These audits were concluded over a period of 6 (six) days.
We hereby confirm that the above information is true and correct:

14th December 2023 – Krugersdorp, South Africa

A handwritten signature in black ink, appearing to read 'Jaarsveld'.

Riaan van Jaarsveld
Security Expert – CCSA / Certified Pen Tester

A handwritten signature in black ink, appearing to read 'Rensburg'.

Lukas Janse van Rensburg
Security Engineer